Hyung Cho

QA Engineer | LLM Agent Evaluation and Root-Cause Debugging | 10 Years of Product Validation

Los Angeles, CA | hcho22@gmail.com | (408) 655-5691 | linkedin.com/in/hyungcho | github.com/hcho22

SUMMARY

Validation engineer with 10 years at Zebra Technologies writing test plans, acceptance criteria, and final reports for 6 shipped hardware products, and isolating failures across hardware, firmware, software, and test infrastructure. Now builds and tests LLM agents, including a support bot with a human handoff and the eval that checks it. Works in Python, SQL, and REST APIs, with checks automated in GitHub Actions.

EXPERIENCE

Independent AI Engineer | Self-directed projects, Remote **Jun 2025 - Present**

Solo product work, built by directing AI coding agents (Claude Code, Codex).

Purvia, permission-aware RAG and support-bot app, open source

- Built a support bot whose draft answers must pass a retrieval gate and LLM faithfulness and completeness judges that fail closed; anything that misses a gate escalates to a human operator queue.
- Designed the escalation eval: an 18-question test set whose deterministic leak checks block pull requests, a weekly LLM-judged check against a 5% false-resolve ceiling, and a check that the offline and runtime judges agree.
- Traced a 27% breach of that ceiling, four weeks running, to the offline judge: every false-resolved row was one the runtime judge had marked not answered, including the reply "I don't have that information."
- After a permissions benchmark published 0.000 for 42 runs because its test users lacked a workspace membership, added guard tests so the evals refuse to report a score when they measured nothing.
- Built a 60-question regression set that scores vector, keyword, and hybrid search and posts the pull-request-versus-main difference on pull requests; it flagged a 0.510 recall@5 drop on a deliberately broken change.
- Wrote access control into the Postgres retrieval SQL so every search runs as the signed-in user, plus a CI job that fails unless a user with no grants gets no protected content back.

Rep Today, native iOS workout app, pre-release

- Grew the iOS unit suite from 829 to 1,687 tests, with 1,665 running on both iOS 18 and iOS 26 in CI, and fixed platform-only failures CI caught, including a layout loop that hung the suite past the 45-minute job limit.

CreativeBridge, K-12 AI story-writing iOS app, internal TestFlight builds

- Built child-privacy controls (age gate, parent consent, PII scrubbing, content moderation, data-retention jobs) and a 29-file test suite mapped to the project's own audit findings, with 48 expected-to-fail tests that keep open gaps visible.

Volunteer AI/ML Engineer | Omdena, Remote **Apr 2023 - Jun 2025**

Unpaid collaborative ML projects with global volunteer teams.

- Fine-tuned open-source LLMs (LLaMA, Falcon) and iterated on chatbot prompts with an evaluation workflow.

Career break and ML retraining **Aug 2018 - Mar 2023**

Self-directed ML study and the FourthBrain ML Engineering bootcamp (2022); 29 public ML projects on GitHub.

Senior Validation Engineer | Zebra Technologies, Agoura Hills, CA **Feb 2008 - Jul 2018**

- Led product validation for 6 shipped hardware products: test plans, procedures, acceptance criteria, and final reports.
- Isolated root causes of failures across hardware, firmware, software, and test infrastructure, and proved the fixes.
- Owned SDK validation across Windows, macOS, and Linux: compatibility, performance benchmarks, and reliability.
- Built C# test automation that replaced manual workflows and cut regression cycles by about 30%.

SKILLS

QA and validation: Test planning, Failure analysis, Regression testing, Test automation, Requirements traceability, Flaky test diagnosis, Quality process

AI evaluation: LLM evaluation, LLM-as-judge, RAGAS, Prompt engineering, Retrieval metrics, LangSmith

Data and APIs: Python, SQL, PostgreSQL, FastAPI, OpenAI API, Anthropic API, Privacy engineering

CI and tooling: GitHub Actions, CI/CD, XCTest, C#, Claude Code

EDUCATION AND TRAINING

B.S., Electrical Engineering, California Polytechnic State University, San Luis Obispo

FourthBrain ML Engineering bootcamp (2022); DeepLearning.AI deep learning and generative AI courses